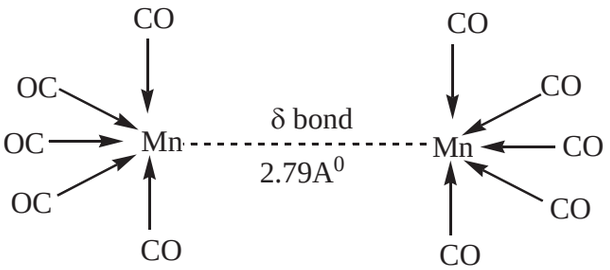
bond, so the formula is more correctly represented as $(CO)_5Mn–Mn(CO)_5$.

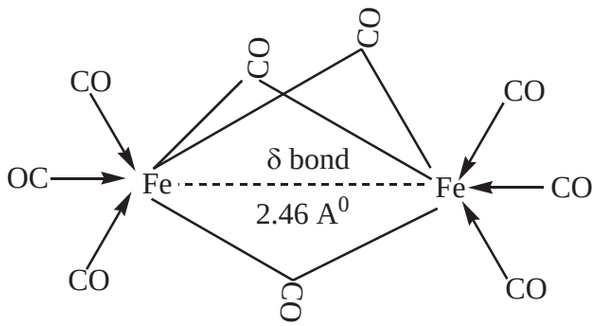


Other examples of this type are,$Tc_2(CO)_{10}$, and $Re_2(CO)_{10}$.

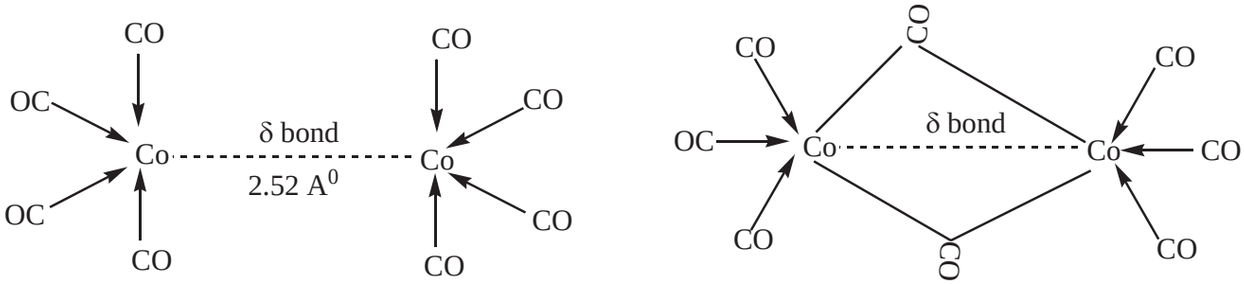
b. Bridged carbonyls:

These metal carbonyls contain one or more bridging carbonyl ligands along with terminal carbonyl ligands and one or more Metal-Metal bonds. For example,

(i) The structure of $Fe_2(CO)_9$, di-ironennea carbonyl molecule consists of three bridging CO ligands, six terminal CO groups and single Fe-Fe Δ bond formed by weak coupling of the unpaired electrons present in two 3d orbitals of two Fe atoms. This bond is represented by dotted line and is called as fractional single bond.



(ii) For dicobaltoctacarbonyl$Co_2(CO)_8$two isomers are possible. The one has a metal–metal bond between the cobalt atoms, and the other has two bridging CO ligands and a metal–metal bond.



Bonding in metal carbonyls

In metal carbonyls, the bond between metal atom and the carbonyl ligand consists of two components. The first component is an electron pair donation from the carbon atom of carbonyl ligand into a vacant d-orbital of central metal atom. This electron pair donation forms $M \xleftarrow{\sigma \text{ bond}} CO$ sigma bond. This sigma bond formation increases the electron density in metal d orbitals and makes the metal electron rich. In order to compensate for this increased electron density, a filled metal d-orbital interacts with the empty π^* orbital on the carbonyl

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